

Toward simplified access to Veluwe ecological geodata through OGC Web Services

MSc Internship Project

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1. Introduction

In recent decades, the volume of ecological data available at the national and international level has grown significantly, with many open-access datasets now accessible through various online platforms (Güntsch et al., 2025). This growth encompasses legacy data (i.e., data from a previous design that can be identified and reused) and datasets from national databases (Chatterjee, 2003). Despite this increased availability, both types of sources present distinct challenges for researchers. Legacy data remains fragmented, dispersed across numerous small-scale studies, making it difficult and time-consuming to identify relevant datasets for a specific region (Reyserhove et al., 2020). Datasets from national databases demand extraction of location-specific information (instead of for the whole country). Ecologists, particularly those involved in data-intensive tasks such as modelling, often face challenges in efficiently locating and manually downloading the data needed to support their research and conduct repeatedly the same preprocessing tasks to adjust the data to their study area.

A useful online resource that can facilitate the task of exploring available data is the standardized Open Geospatial Consortium (OGC) web services (OGC, 2016). These services contribute to a more efficient discovery, filtering, and retrieval of geospatial data directly from online sources. These services offer a powerful way to automate data access, but they remain unfamiliar to many ecologists and can be difficult to use without technical expertise. This project addresses that gap by developing a data pipeline for the discovery and processing of ecological datasets from multiple sources, with a focus on the Veluwe region – one of the sites involved in the *Netherlands Long-term Ecosystem Research Network* (LTER-NL). The development of this data pipeline for the Veluwe is intended to directly contribute to LTER-NL's mission of making long-term ecological data more discoverable, accessible, and usable for researchers.

The referred pipeline aligns well with the FAIR principles (Findable, Accessible, Interoperable, Reusable), which guide good data management and stewardship (Wilkinson et al., 2016). The first part aligns with the "Findable" principle by making geospatial datasets easily searchable through machine-readable catalog services. The second and fourth parts of the pipeline support the "Accessible" principle by relying on standardized protocols (WCS and WFS) to retrieve dataset details and to download data through well-documented, open interfaces. The pipeline also ensures "Interoperability" by using OGC standards like CSW, WFS, and WCS. These standards facilitate the exchange of data and metadata in structured formats (e.g., GML, XML, GeoTIFF), allowing integration across different platforms and tools. Finally, the extraction of technical metadata (e.g., native spatial resolution and geometry types) facilitates "Reusability" by allowing users to assess whether a dataset meets their needs, while also supporting proper attribution, licensing, and provenance tracking.

The aim of this project is to develop an automated and reproducible pipeline that identifies, filters, and selects relevant ecological geospatial datasets for the Veluwe region using OGC Web Services. In addition to data acquisition, the pipeline also extracts metadata fields from the *NGR* to assess their alignment with the metadata structure used in the LTER-LIFE project. By comparing the availability and compatibility of these fields, the project contributes to understanding how existing national geospatial infrastructures can support the metadata needs of long-term ecological research.

2. Research questions

To guide this development, the following research questions are formulated:

RQ1: How can we create a user-friendly workflow to select and retrieve data from different data sources?

It is hypothesized that only a small fraction of the available datasets will be retrieved, due to issues such as incomplete metadata and lack of OGC-compliant services. To explore this, scripts will be developed to access metadata through OGC protocols (CSW, WFS, WMS, and especially WCS), applying spatial filters and metadata-based criteria to select datasets relevant to the Veluwe region.

RQ2: To what extent do the metadata fields defined for LTER-LIFE have equivalents in the *NGR* catalogue, and how frequently are these fields present in existing *NGR* metadata records?

It is expected that a subset of the metadata fields defined for LTER-LIFE will have direct equivalent in the *NGR* catalogue, particularly those aligned with common geospatial metadata standards such as ISO 19115. However, not all LTER-LIFE-specific fields, especially those tailored to long-term ecological monitoring, are expected to be present. The frequency of matching fields in *NGR* metadata records is likely to vary depending on the dataset's provider.

3. Study area

The area of interest is shown in **Figure 1**, which presents the outline of the Netherlands, and highlights the focus area of this study: the Veluwe region. The Veluwe is the largest continuous forested and natural area in the lowlands of northwestern Europe, located in the province of Gelderland in the eastern Netherlands (van der Heide et al., 2008). Its central landscape features a ridge of hills forming part of the Dutch sand region, where fine periglacial sands deposited during the last Ice Age (Weichselian, 100,000 to 10,000 years ago) predominate (van der Heide et al., 2008). The natural landscape consists of a mix of forests, heathlands, sand drifts, lakes, and moorlands (LTER-NL, n.d.).

Ecologically, the Veluwe is of exceptional importance due to both its considerable size, covering approximately 1,000 square kilometres, and its ecological quality (van der Heide et al., 2008). Furthermore, the unique combination of densely populated areas and natural landscapes in the Veluwe offers valuable opportunities to study the complex interactions between humans and nature (LTER-NL,



Figure 1 | Map of the Netherlands with white outline, and Veluwe region with orange outline (left); detail of Veluwe region showing the combination of forests, agricultural areas and urban areas (right).

n.d.). Besides the ecological importance, the Veluwe area is also a primary strategic groundwater system in The Netherlands and an important region for drinking water supply (Huijgevoort et al., 2020).

These characteristics make the Veluwe an important region for multidisciplinary research, requiring access to extensive and diverse geospatial datasets. The boundary depicted in **Figure 1** illustrates one example of how the Veluwe can be delineated and will be adopted for this study. However, the spatial extent of the region remains flexible and can be redefined by the user within the pipeline.

4. Methodology

4.1. Overview of the pipeline

The pipeline consists of four main components (**Figure 2**). The first is focused on harvesting metadata of available geodata using the *Catalog Services for the Web* (CSW) from the *Nationaal Georegister* (NGR). NGR is an important gateway for discovering geoinformation in the Netherlands (Nationaal Georegister, n.d.). The second part uses *OGC web service* to extract additional details important to write the data request, such as native spatial resolution, total number of features per record and type of geometry. The third block is only about filtering by location (spatial filter) and by content (content-based filter). In the final step, the selected raster and vector layers are retrieved and downloaded via WCS (Web Coverage Service) and WFS (Web Feature Service) protocols. The following sections provide detailed explanations of each step.

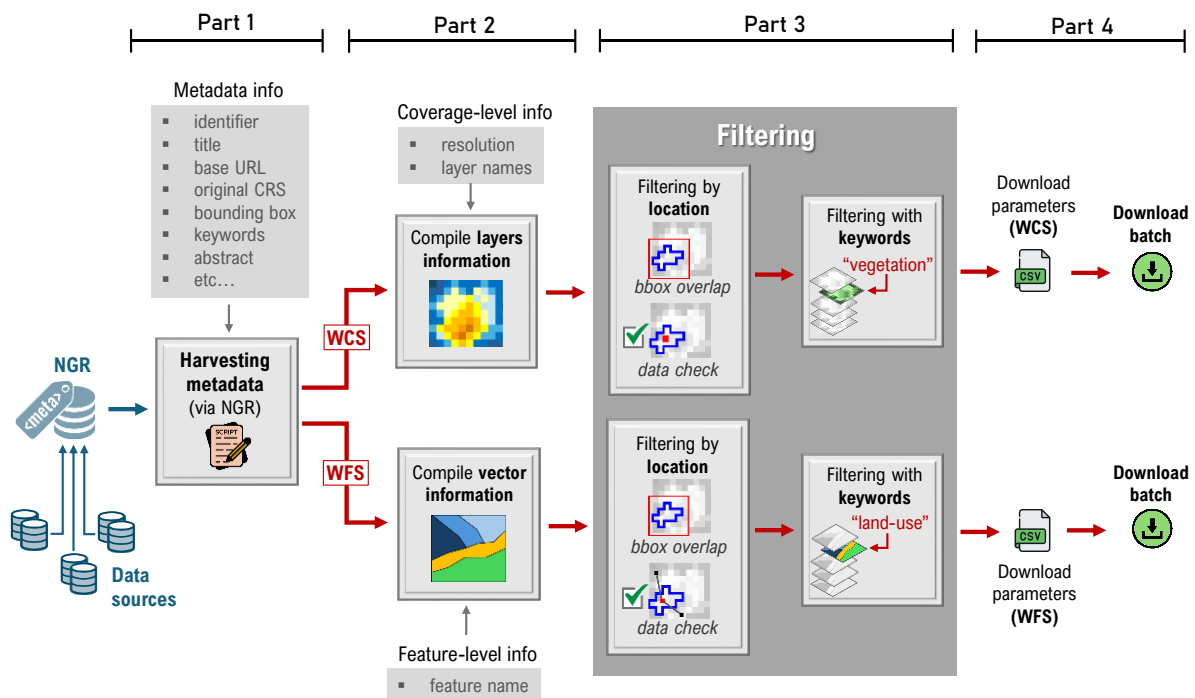


Figure 2 | Overview of the data pipeline workflow split into four parts: (1) collect all metadata in National Georegister (NGR), which centre data from diverse sources; (2) Addition of important information, such as spatial resolution and layer names, to help writing the data request; (3) spatial and content type filters; and (4) compilation of selected data and download.

4.2. Harvesting metadata from NGR (Part 1)

The first task involved creating scripts that automatically extract metadata records from a Catalogue Service for the Web (CSW) endpoint using a structured, XML-based request-and-response loop. The

scripts were written in Python (version 3.11.9) and run in Jupyter Notebooks (version 6.5.4), which provide an interactive environment for code and analysis. The *requests* package was used to send queries to the metadata server in the form of XML messages. To make sense of the XML responses, the script contains the *xml.etree.ElementTree* package to read and interpret (or "translate") the returned information in a structured way.

This study focused only on the catalogue containing all metadata from the Nationaal Georegister (NGR) (<https://www.nationaalgeoregister.nl>), which is referred here as CSW metadata. NGR serves as an important gateway for discovering geoinformation in the Netherlands (Nationaal Georegister, n.d.). NGR hosts a catalogue of metadata records from major data originators across the country, such as PDOK (Publieke Dienstverlening Op de Kaart), RIVM (Rijksinstituut voor Volksgezondheid en Milieu), and RWS (Rijkswaterstaat) (Error! Reference source not found.).

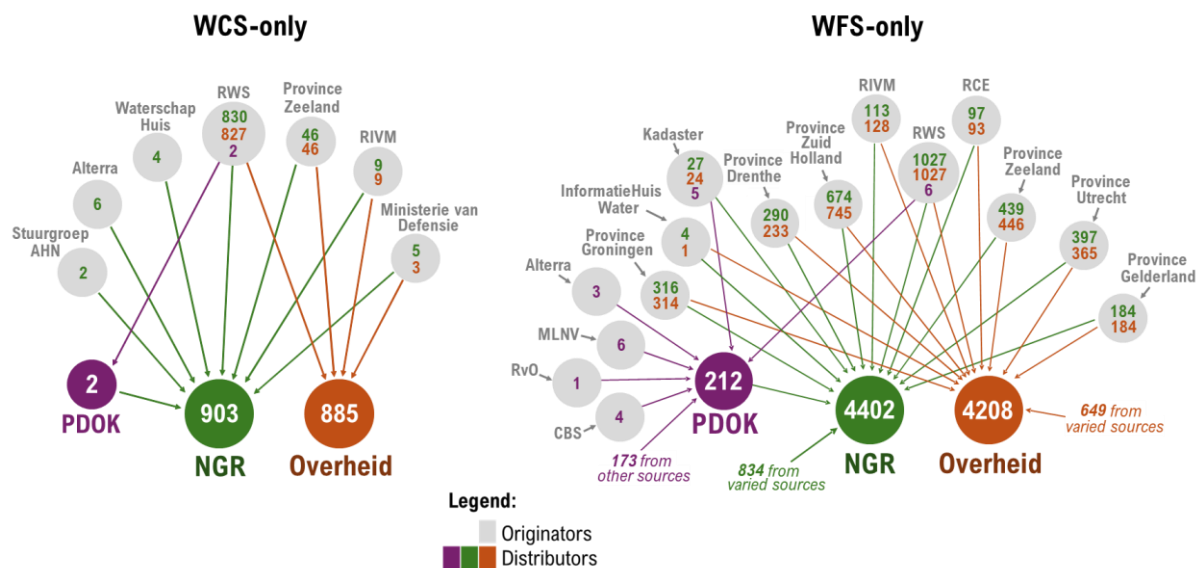


Figure 3 | Relationships between geodata originators and distributors in the Netherlands (illustrative, not exhaustive). NGR and other distributors collect metadata from various national geodata providers. Coloured numbers inside grey circles indicate how many datasets each originator provides to a distributor. The numbers inside the distributors' circles represent the total datasets they offer. Some datasets may appear in multiple distributors, but this figure does not show those overlaps. Left plot is related to WCS (i.e., service for rasters), and right plot is related to WFS (i.e., service for vectors).

To retrieve metadata records, the *GetRecords* operation was used, implemented via the "owslib.csw" package to structure the CSW requests as follows in **Box 1**:

```
csw = CatalogueServiceWeb("https://www.nationaalgeoregister.nl/geonetwork/srv/dut/csw")
csw.getrecords2(maxrecords=100, startposition=1, esn="brief")
```

Box 1 | Example of a basic CSW *GetRecords* request using OWSLib to retrieve the first 100 metadata records from NGR.

Initially, a total of 20 metadata fields (**Appendix A1**) were selected based on the LTER-LIFE metadata fields (**Appendix A2**), prepared, organized and shared for this study by the researchers involved in the project developing the LTER-LIFE infrastructure. Unfortunately, not all metadata fields that were equivalent to the 23 metadata fields prepared by LTER-LIFE were found, mainly due to differences metadata standards. LTER-LIFE includes 17 fields based on DCAT-AP and 5 fields based on ISO 19115. In contrast, NGR primarily uses ISO 19115 for most of its metadata, with a smaller portion following ISO 19110.

To allow a better organization of the pipeline, two other fields were incorporated: “Metadata Standards” and “OGC Web Service”. Also, the “Distribution access URL” field was split into “wcs_GetCapabilities_url” and “wfs_GetCapabilities_url”.

4.3. OGC service metadata operations (Part 2)

The second part of the pipeline uses the information retrieved from the initial metadata to access a deeper level of detail through *OGC web service*, such as **GetCapabilities**, **DescribeCoverage**, and **DescribeFeatureType**. The *GetCapabilities* operation is used by both WCS and WFS to retrieve a list of available datasets on the server, along with information about the supported operations and request parameters (GeoServer Project, n.d.). The *DescribeCoverage* operation is specific to WCS. It returns an XML document that provides detailed metadata about one or more coverages, including spatial extent, resolution, and coordinate reference systems (GeoServer Project, n.d.). The *DescribeFeatureType* operation is specific to WFS. It returns an XML document that defines the internal structure of a feature type, including the names and data types of all attributes, as well as the geometry type and any constraints such as whether fields are optional or required (Geoserver Project, n.d.). These operations were accessed by appending specific parameters to the service’s base URL, as in **Box 2**.

```
GetCapabilities for WCS: [base URL]?service=WCS&request=GetCapabilities&version=[version]
GetCapabilities for WFS: [base URL]?service=WFS&request=GetCapabilities&version=[version]
DescribeCoverage: [base URL]?service=WCS&request=DescribeCoverage&version=[version]&coverageId=[coverage_name]
DescribeFeatureType: [base URL]?service=WFS&request=DescribeFeatureType&version=[version]&typeName=[feature_type_name]
```

Box 2 | Examples of metadata operations in OGC services.

4.3.1. Coverage names

To download data via WCS, the coverage name is required. Sometimes, this name can be found in the CSW metadata under the `<gmd:CI_OnlineResource>` tag. However, this information is not always present there. As an alternative, coverage names can be retrieved directly from the WCS *GetCapabilities* response. The location of the coverage name in *GetCapabilities* depends on the WCS version:

- In WCS 2.0.x, coverage names are listed under `<wcs:CoverageSummary>` using the `<wcs:CoverageId>` tag.
- In WCS 1.0.x, they are found within `<wcs:CoverageOfferingBrief>` using the `<wcs:name>` tag.

To handle these version differences, a fallback approach is used: the script first tries the structure of WCS 2.0.x, and if no names are found, it automatically retries using the WCS 1.0.x structure. This workaround effectively resolves the version-related inconsistency.

However, another issue arises with naming format differences. In CSW metadata, coverage names often appear in a namespace:layer format (e.g., `ahn:ahn3_05m_dsm`), while in *GetCapabilities*, the same layer might be named with a double underscore, like `ahn__ahn3_05m_dsm`. Since the format from *GetCapabilities* is the one accepted in actual WCS operations, it is preferred as the "official" layer name.

Initially, the solution seems simple: just use the name from *GetCapabilities*. However, in some cases, a single *GetCapabilities* URL serves many different datasets (or CSW identifiers), resulting in a long list of coverages. To identify the correct one, a comparison is needed. Because the names in CSW and WCS use different separators, a simplified matching is performed: both names are "cleaned" by extracting only the part after the separator (e.g., `ahn3_05m_dsm`), and these cleaned names are compared. Once

a match is found, the original name from *GetCapabilities* (with double underscore) is selected and used for subsequent WCS operations.

4.3.2. Native spatial resolution

To determine the native resolution of a raster layer, the information must be retrieved from the *DescribeCoverage* operation.

In WCS version 2.0.1, the resolution is provided directly as vectors, found in the `<gml:offsetVector>` elements. These vectors describe the cell size in each direction.

In WCS version 1.0.0, the resolution must be calculated. The `<gml:low>` and `<gml:high>` elements define the pixel-based grid boundaries — the starting and ending positions of the raster grid. From these values, the total number of pixels in the X and Y directions can be computed.

To relate this grid to real-world dimensions, the `<gml:pos>` elements are used:

- `<gml:pos[1]>` provides the coordinates (in map units, such as meters) of the lower-left corner
- `<gml:pos[2]>` gives the upper-right corner

By dividing the total distance (in meters) between these corners by the number of pixels, the X and Y resolution (in meters per pixel) can be calculated.

4.3.3. Feature names

The WFS feature name is retrieved using a two-step fallback strategy. It first tries to extract the feature name directly from the CSW metadata by parsing the XML structure under the `gmd:CI_OnlineResource` element. If a valid WFS protocol is detected, the associated `gmd:name` is collected as the potential feature name. If no feature name is found in the CSW metadata, the code sends a live *GetCapabilities* request to the WFS service URL. It parses the returned XML to extract all available `<wfs:FeatureType>` names.

4.4. Spatial and content-based filtering (Part 3)

4.4.1. Spatial filtering

The third stage of the pipeline focuses on filtering out spatially irrelevant data. The first spatial filter was based on the geographic extent of the study area: the Veluwe region (**Figure 1**). Using its outline as a reference, each dataset's bounding box (retrieved from the metadata) was tested for spatial intersection. This required reprojecting all bounding boxes to EPSG:28992, which is the coordinate reference system used for the Veluwe outline. The intersection check was carried out using the *Shapely* library's *intersects()* method.

However, initial inspection revealed that the bounding boxes provided in the metadata are not always reliable indicators of actual data coverage. For example, some datasets cover a small area in the Netherlands, but the correspondent bounding box covers the entire national territory. This issue led to false positives during spatial filtering in Veluwe, for example. Therefore, the pipeline includes a second, more precise round of spatial filtering, especially for raster datasets.

In this second filter step, a set of points was generated at regular intervals inside the Veluwe outline. Around each point, a very small bounding box was created based on the native spatial resolution of the raster. These small bounding boxes were then used to individually query the datasets, checking whether any data was actually present at those locations. If at least one of these queries resulted in a successful

data retrieval (i.e. the service began returning data for that area), the dataset was considered to have valid coverage within the Veluwe region (**Figure 4**).

Although this method significantly increased processing time, it provided a much more accurate assessment of whether a dataset truly intersected the study area. In some cases, a dataset's bounding box may extend far beyond the Veluwe, causing the pipeline to attempt downloading large tiles that include the region but contain no valid data (e.g. only NaN values). By performing this second-level spatial filter, the pipeline avoids unnecessary downloads from such tiles with no useful information.

Identifying spatial coverage, in the broad sense, is very important because there are size limits per request for downloading raster data via OGC services. This constraint becomes especially problematic for high-resolution datasets (e.g., 0.05 m or 5 cm resolution, such as aerial imagery). In those cases, even when using tiling strategies (explained in **section 4.5.1.3.**), the number of required tiles can become unmanageably large.

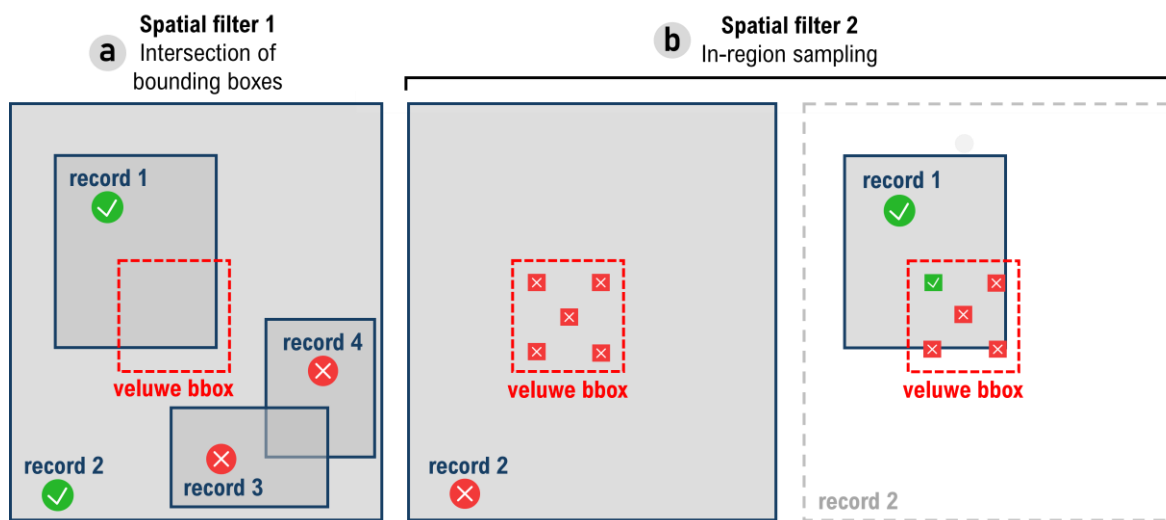


Figure 4 | Overview of spatial filters: (a) spatial filter 1 is based merely in the intersection between the bounding box from the study area (Veluwe) and the bounding boxes from all the metadata records; (b) spatial filter 2 is conducted by spread a few very small bounding boxes, referred here as samples, inside the study area and test a process of downloading in each of them. If all samples return empty, it is considered as not containing data and is filtered out. If at least one sample start to download data, it is interrupted, and the respective record is filtered in.

4.4.2. Content-based filtering

Once datasets intersecting the Veluwe region have been identified, it is useful to further restrict the selection based on content relevance. A content-based filtering step can be applied to retain only those datasets that are potentially relevant to ecological studies, for example. This is achieved by comparing a user-defined list of thematic keywords with the keywords extracted from each dataset's metadata.

Since the keywords metadata field is sometimes missing or contain only a few words, it was enriched by extracting additional terms from the abstract metadata field. These words were appended to the original keywords. All resulting keywords were converted to lowercase to facilitate matching with the user-defined word list.

To handle cases where exact matches are unavailable due to spelling variations, formatting inconsistencies, or incomplete metadata, *approximate string matching*, also known as *fuzzy string matching*, was used. This method calculates similarity scores between strings based on distance algorithms, such as *Levenshtein* distance (Pikies & Ali, 2021). In this study, fuzzy matching was

implemented using the *FuzzyWuzzy* Python package (SeatGeek, n.d.). A minimum fuzzy similarity threshold of 80 was used by default, though the user can adjust this value as needed.

The list of keywords used for fuzzy matching was compiled in consultation with NIOO colleagues and is intended to reflect themes of ecological interest, including land-use, roads, recreative areas, vegetation maps, etc. It also serves as a test case for validating the pipeline. The list included the following terms:

```
["landgebruik", "paden", "recreatievoorzieningen", "recreatie", "wegen", "barrières", "recreatiedichtheid",  
"leefgebieden", "wildbeheer", "terrein eigenaars", "eigenaars", "roodwild", "tellingen", "wolf",  
"vegetatiekaarten", "vegetatie", "Recreatiezoneringsmaatregelen"]
```

4.5. Data download (Part 4)

Once all datasets have been discovered and filtered, the necessary information to request them is stored in two CSV files: one for raster data (WCS) and the other for vector data (WFS). To manage batch downloads, each file includes a column named “*download*”, which is initially set to “*no*” for all entries. The user is expected to manually review the files and change the download value to yes for the datasets they wish to include in the final download step of the pipeline.

4.5.1 Rasters via WCS

To download rasters, it was used the *GetCoverage* operation. In **Box 3**, an example of a data request using WCS version 2.0.1. However, differences in parameters across WCS versions, along with server-side memory limitations for data transfer, present challenges for generalizing this request format. These issues required several workarounds, which are explained in the following sections.

```
service=WCS&  
version=2.0.1&  
request=GetCoverage&  
coverageId=layer&  
subset=E(12345,5)&  
subset=N(45678,9)&  
subsettingCrs=EPSG:28992&  
outputCrs=EPSG:4326&  
format=image/tiff
```

Box 3 | Example of a data request using WCS version 2.0.1

4.5.1.1. WCS versions

To access correctly the parameters to request download, it is important to know which WCS version (e.g., 1.0.0, 2.0.1) is supported by the server of a specific endpoint. A function was implemented to detect WCS version. It receives the URL of the *GetCapabilities* and tries to access the attribute “*version*”. Not always the version will be available there, so it does not return any information, the code tries to get it from the namespace indicators (e.g., “<http://www.opengis.net/wcs/2.0>” represents version “2.0.1”). In case all of these fail, it returns the version 1.0.0., which is the older version and possibly did not need to specify the version when there were no other versions available yet.

The version WCS 2.0.1 uses the properties “*scaleSize*” and “*scaleSize_y*”, which represent the number of pixels in the horizontal and vertical axis of the raster, respectively. These values are obtained by calculating the total distance of the longitude (and latitude) divided by the native resolution (i.e., $\text{scaleSize} = (\text{xmax} - \text{xmin}) / \text{resx}$). The version WCS 1.0.0. uses the properties “*width*” and “*height*”, which

represent the same idea of “scaleSize”. Additionally, this version also accepts directly `resx/resy`, which is the direct value of the spatial resolution (meters per pixel).

It was decided to retrieve the raster with native resolution rather than offer the option to spatially aggregate by asking resampling to the server. Even though the aggregation could speed up the download, not to do it was preferred in order to prioritize the original data integrity as much as possible.

Other parameters beyond resolution also differ between WCS versions. Compared to version 2.0.1 (as shown above), in version 1.0.0:

- “coverageId” becomes simply “coverage”,
- “subset” is replaced by “BBOX”, and
- “subsettingCrs” and “outputCrs” are combined into a single “CRS” parameter used for both.

4.5.1.2. Axis labels

The way coordinate axis labels are defined in the “DescribeCoverage” response can introduce complications when constructing valid download requests. In the example above, the “subset” parameter uses axis labels “E” and “N”, but variations across datasets in the NGR include labels like `Easting / Northing`, `Lat / Long`, or `X / Y`. To handle these inconsistencies, a function was implemented to detect the correct axis labels. It sends a “DescribeCoverage” request to the WCS endpoint, parses the XML response, and extracts the “axisLabels” attribute from the `<gml:Envelope>` element. If this process fails, it defaults to using “x” and “y” as fallback axis names.

4.5.1.3. Tiling and merging

WCS servers often have memory limits that restrict how much data can be delivered in a single request. These limits vary across servers, and calculating the exact transferable data size can be complex. To work around this, the chosen strategy involves splitting the original raster into smaller tiles, each within a size that the server can handle. Once all tiles are downloaded, they are merged locally to reconstruct the full raster, as close as possible to the original. This process is not handled by OGC services and must be implemented on the client side.

The tile size still needs to be defined. Various square tile sizes were tested on different datasets (not documented in this report), and the largest tile size that consistently avoided server-side errors was 4,000 × 4,000 pixels. This size was set as the default in the code, though it can be adjusted by the user.

The tiling function takes as input the bounding box of the study area and the spatial resolution. Together with the default tile size (4,000 × 4,000), this is sufficient to compute the number of tiles required. The outline of the study area is an additional input used to filter out tiles falling outside the area of interest, thereby reducing unnecessary downloads.

4.5.2. Vectors via WFS

To download vector data, the *GetFeature* operation of the WFS service was used. An example of such a request, using WFS version 2.0.0, is shown in **Box 4**. Before downloading the data, the total number of features in the layer is requested from the server (details in **section 4.5.2.1**). The request also includes the bounding box of the study area to ensure that only features within this area are returned. Because the server may limit how many features can be returned at once, the data is downloaded in chunks, using the *startIndex* and *maxFeatures* parameters for pagination. Each chunk is first downloaded in

GeoJSON format and then loaded into a GeoDataFrame. Finally, all chunks are combined into a single full GeoDataFrame containing all the features.

```
service=WFS&
version=2.0.0&
request=GetFeature&
typename=layer&
bbox=12345,45678,12350,45687,EPSG:28992&
srsName=EPSG:4326&
outputFormat=application/json
```

Box 4 | Example of a data request using WFS version 2.0.0.

4.5.2.1. Feature count

This step asks the WFS server how many features are available in a specific layer, without actually downloading them. It does this by sending a special *GetFeature* request with `resultType=hits`, which tells the server to return just the feature count. Since different WFS versions use slightly different formats, it is tried first the WFS version 1.1.0, where the key is `typeName`. If this version does not work, then it tries WFS version 2.0.0, where the key is `typename`. If both request fails, it will retry up to 5 times, waiting a few seconds between each try. Once a valid response is received, the code reads the XML and looks for `numberOfFeatures` in version 1.1.0, or `numberMatched` in version 2.0.0. Knowing how many (individual) features is within the feature layer is a necessary information to allow the download process to complete correctly.

5. Results and discussion

5.1. Metadata overview

In total, 9,238 metadata records were retrieved from NGR at 17/07/2025 in 1h and 2min, at an average rate of about 2.5 records per second. Only the fields “Unique Identifier” and “Metadata Standard Information” contained valid values in all records. For the other 21 metadata fields, most fields (e.g., Metadata date) were partially filled, while some fields (e.g., Data publisher) were missing in most records (**Figure 5**).

A total of 8,718 metadata records conform to the ISO 19115 standard, while 638 records follow the ISO 19110 standard, which specifically describes *Feature Catalogues* (**Figure 6a**). Regarding resource types, 7,626 records correspond to datasets, 938 to services, 66 to GIS datasets, and 14 to series (**Figure 6b**). Among the OGC services represented in the metadata, WMS (Web Map Service) is the most prevalent, associated with 5,946 records, followed by WFS with 4,480, and WCS with 928. Less commonly, WMTS (Web Map Tile Service) appears in 100 records, while Sensor Observation Service (SOS) and CSW are represented with only 9 and 3 records, respectively (**Figure 6c**).

For metadata records associated with WCS, *Rijkswaterstaat* is the primary provider, accounting for 857 records (94%). The remaining WCS records are distributed among *Provincie Zeeland* (45 records), *RIVM* (9 records), and the *Ministerie van Defensie* (4 records) (**Figure 7a**).

With respect to WFS services, there are a total of 70 responsible parties. *Rijkswaterstaat* again leads, contributing 1,032 records. For visualization purposes, all responsible parties with 20 records or fewer were grouped into a consolidated category labelled “others” (**Figure 7b**).

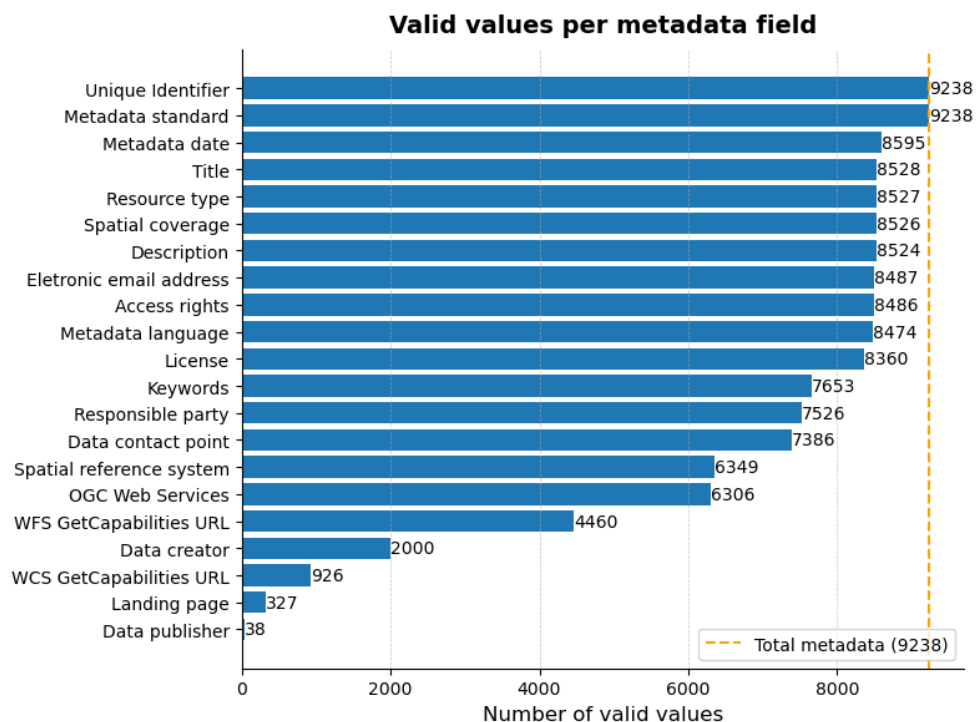


Figure 5 | Summary of all valid values returned to the metadata fields of the 9238 records from NGR.

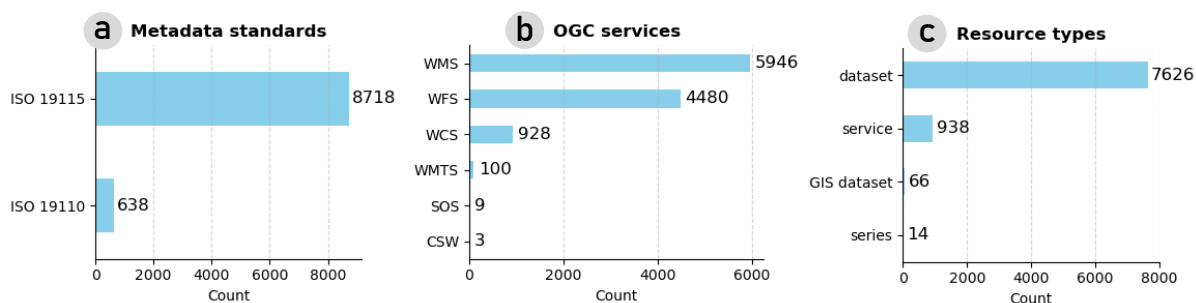


Figure 6 | Summary of metadata fields corresponding to those used in LTER-LIFE: (a) Total count of retrieved metadata standards; (b) total count of resources stratified per types; (c) total count of OGC service types offered by each record's endpoint. Note that a single resource may offer multiple OGC services.

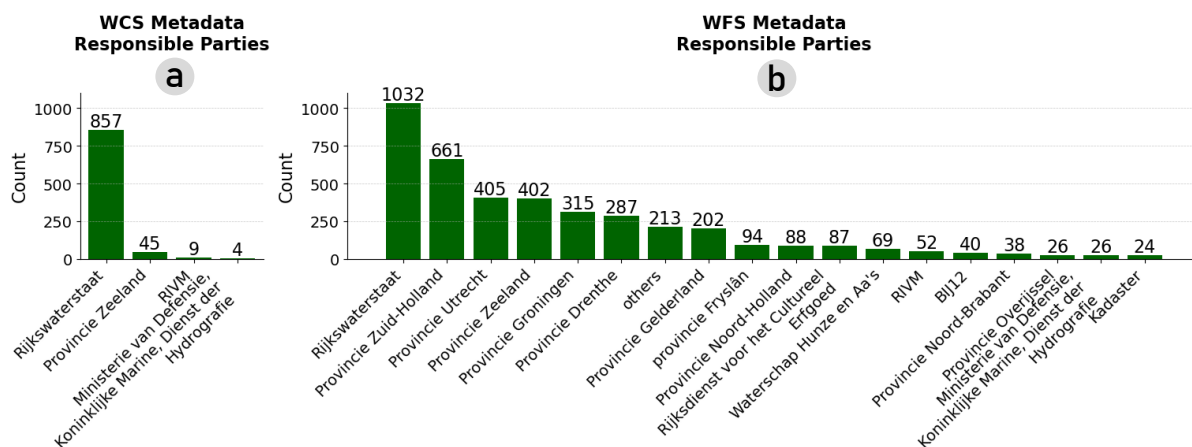


Figure 7 | Total counts of companies responsible for the WCS (a) and WFS (b) metadata records in NGR.

5.2. Spatial filtering results

For WCS, the catalogue initially contained 912 records (after removing duplicated layers). A first-pass intersection test between each record's bounding box and the Veluwe outline (spatial filter 1) reduced the set to 658 entries. A more detailed check then sampled a small patch within the Veluwe for each of those 658 coverages to verify that data were actually present (spatial filter 2). This refinement cut the list to 28 viable coverages but required 23 minutes to finish, averaging around 1.7 iterations per second (Figure 8).

For WFS, the starting catalogue contained 4,215 metadata records, of which 20 lacked bounding-box information and were excluded from spatial processing. Applying spatial filter 1 yielded 2,001 intersecting records in well under one second. Spatial filter 2 then performed the same in-region sampling used for WCS; it retained 1,289 feature layers and took 17 minutes to run, corresponding to about 2.3 iterations per second (Figure 9).

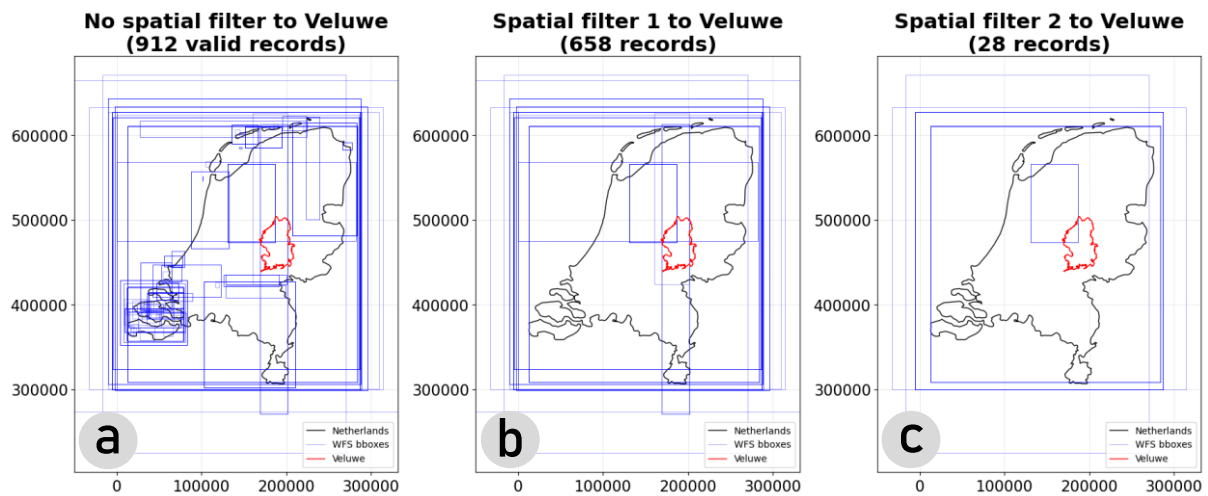


Figure 8 | (a) Bounding boxes (blue) for all 912 original WCS records; (b) 658 bounding boxes remaining after applying the first spatial filter, which removed records that did not intersect the Veluwe outline (shown in red); (c) Final spatial filter based on sampling within the Veluwe to verify the presence of actual data. Only records with confirmed data were retained, resulting in 28 remaining entries.

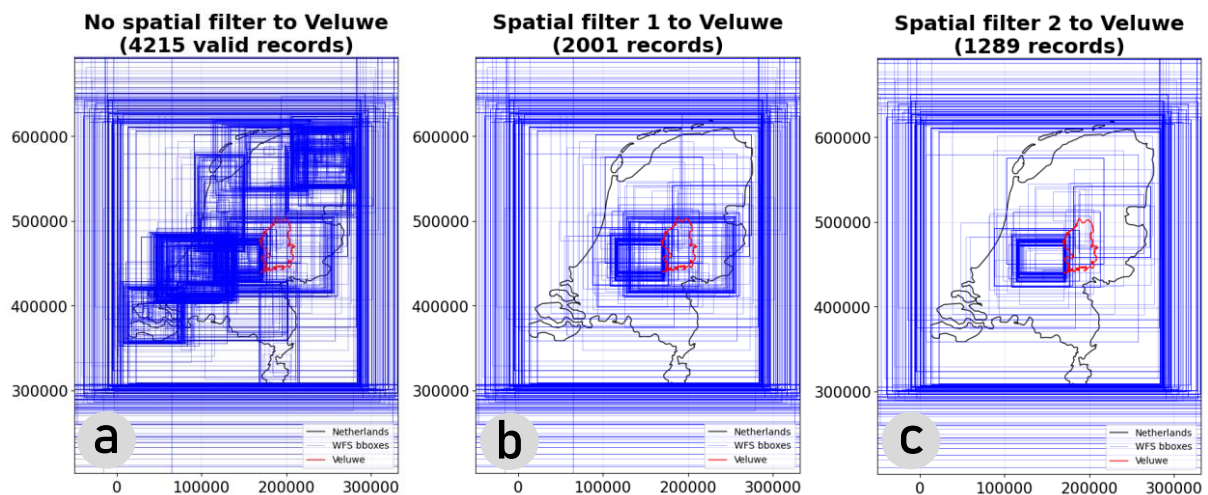


Figure 9 | (a) Bounding boxes (blue) for all 4,215 original WFS records; (b) 2001 bounding boxes remaining after applying the first spatial filter, which removed records that did not intersect the Veluwe outline (shown in red); (c) the final spatial filter was based on issuing a GetFeature request to count the number of features within the Veluwe bounding box. Only records that returned more than one feature were retained, resulting in 1,289 remaining entries.

5.3. Content filtering results

Fuzzy matching between the words from the *wish list* and the keywords from the WCS records completed in under one second, returning 11 entries. To give a clearer picture of how the content of the dataset is filtered, **Figure 10** and **Figure 11** show the number of keyword entries for WCS and WFS, respectively, after each step: from the original data, through spatial filters, and finally after applying the content filter. In all three cases, missing keywords make up the majority of entries¹. Even when keywords are present, they are sometimes not very informative. Examples include terms like “31105748”, “*Flora en Fauna*”, or “*Natuur en Milieu*”, which give little insight into the actual content.

At first glance, the keyword bar plot for the content-based filter (**Figure 10c**) suggests a certain degree

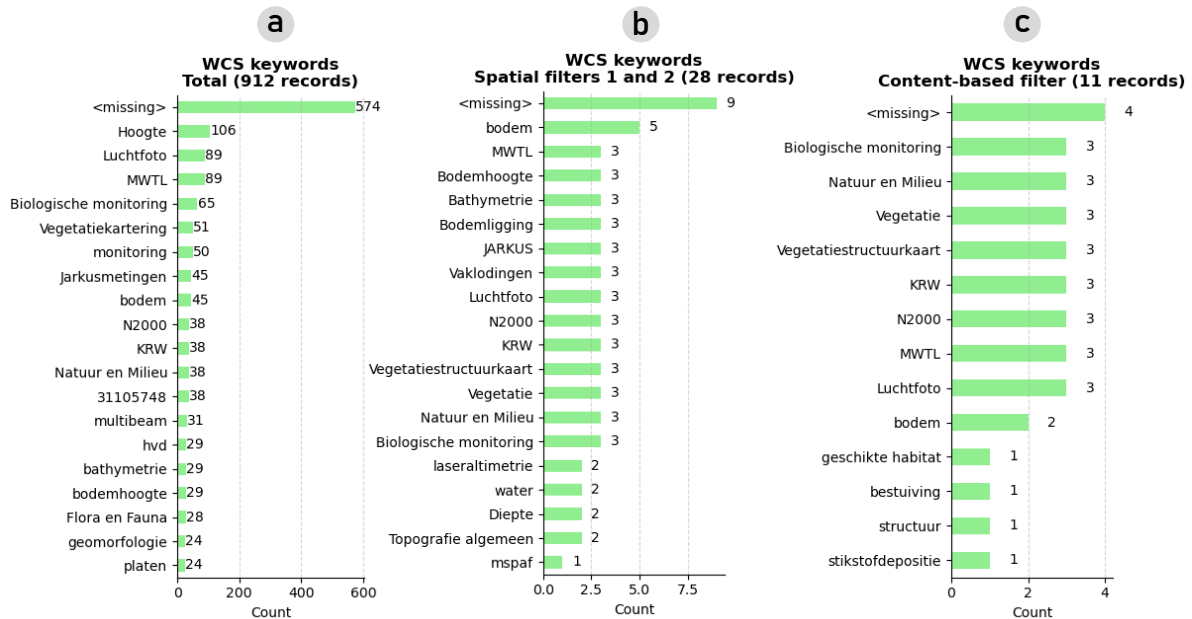


Figure 10 | Counts of resulting keywords before (a) and after the spatial filters (b) and content-based filter (c) for WCS.

of content diversity, with terms related to vegetation, soil ('*bodem*'), and biological monitoring. However, most of these keywords come from just three data records related to the Orthophoto mosaic: *Luchtfoto*, *MWTL*, *N2000*, *KRW*, *Vegetatiestructuurkaart*, *Vegetatie*, *Natuur en Milieu*, and *Biologische monitoring*. Among the remaining eight records, four have no keywords (missing), while the other four contain only a few terms: *bestuiving*, *geschikte habitat*, *bodem*, *structuur*, and *stikstofdepositie*. This indicates that the apparent keyword diversity is limited to a small number of records and highlights the difficulty of extracting meaningful information from keywords, as many are either missing or concentrated in a few datasets.

Specifically for WCS, since the number of records was very low, the content-based filter was not applied for downloading. Instead, the results from the previous spatial filtering step were retained as they were, totalling 28.

For WFS records, the fuzzy matching yielded 502 matches and took 16 seconds to complete. Similarly to WCS records, missing keywords also made up the majority of WFS entries. The number of records

¹ As a reminder, the content-based filter was applied not only to keywords but also to words found in the abstract. This explains why some records with missing keywords were still included in the filtered list.

with missing keywords remained relatively high across all three cases, and less informative terms like 'Downloadable Data', 'NATUUR EN LANDSCHAP', and 'Informatief' were also present. The content-based filter (**Figure 11c**) primarily captured records related to the transport network and roads, as these keywords appeared among the top six most frequent. This suggests a large overlap of data with redundant types of information.

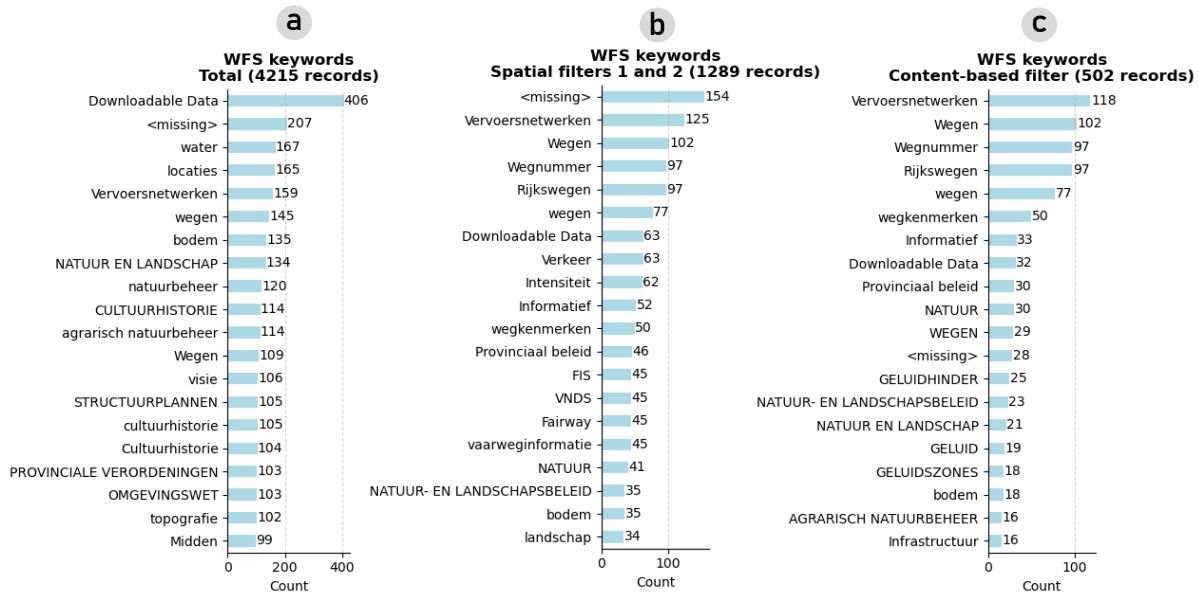


Figure 11 | Counts of resulting keywords before (a) and after the spatial filters (b) and content-based filter (c) for WFS.

5.4. Successful downloads

5.4.1. WCS results

Out of a total of 28 raster datasets, 25 were successfully downloaded (**Table 1**). Among these, 19 rasters had a spatial resolution coarser than 18 meters, which allowed them to be downloaded in a single request without the need for tiling. This group amounted to 364 MB and took approximately 3 minutes to download. Three rasters, with moderate to fine resolutions between 5 and 13 meters, required tiling (ranging from 2 to 8 tiles each) and totalled 621 MB. These were downloaded in 3 minutes and 43 seconds. The remaining three rasters had a much finer resolution of 0.5 meters and required between 221 and 493 tiles. These downloads were more time-intensive, taking over 5 hours and totalling 99 GB. Examples of these downloaded rasters can be found in **Appendix A3**.

The remaining 3 rasters are aerial photo layers technically available for download, but, as these have extremely high resolutions (0.25 and 0.05m), it greatly increases the amount of data to be transferred. Downloading them required to split the request into 1826 or more tiles, and the download was tried but just the first aerial photo was taking more than 6h and was cancelled due to impracticality.

These high-resolution data become even more challenging to download because the metadata only provides a general bounding box that covers the entire country, rather than a precise bounding box for the actual data. Even splitting only within the Veluwe region, numerous unnecessary tiles are requested, making the process intractable.

Table 1 | Summary of downloaded WCS records.

i	Coverage name	Spatial resolution (m)	Success	Tiles	Duration (s)	Size (MB)
1	ank_rivm_r1_gm_soilstr	(25.0, 25.0)	TRUE	1	16.39	25.51
2	ank_rivm_r13_gm_afdekking	(25.0, 25.0)	TRUE	1	4.62	6.53
3	ank_rivm_r19_gm_n-depositie	(1000.0, 1000.0)	TRUE	1	1.68	0.03
4	ank_rivm_r3_gm_diseasesupp	(25.0, 25.0)	TRUE	1	16.47	25.51
5	ank_rivm_r4_gm_waterreg	(25.0, 25.0)	TRUE	1	17.47	25.51
6	ank_rivm_r5_gm_naturalatt	(25.0, 25.0)	TRUE	1	28.46	25.51
7	ank_rivm_r60_gm_geschiktehabitatbestuivers	(100.0, 100.0)	TRUE	1	2.1	1.59
8	ank_rivm_r9_gm_mspaf	(1000.0, 1000.0)	TRUE	1	1.08	0.03
9	bodemhoogte_20mtr__bodemhoogte_20mtr_2022_hillshade	(20.0, 20.0)	TRUE	1	12.75	40.80
10	bodemhoogte_20mtr__bodemhoogte_20mtr_2023_hillshade	(20.0, 20.0)	TRUE	1	13.12	40.80
11	bodemhoogte_20mtr__bodemhoogte_20mtr_hillshade	(20.0, 20.0)	TRUE	1	14.6	39.84
12	bodemhoogte_ijsselmeergebied__bodemhoogte_ijg_2006	(5.0, 5.0)	TRUE	8	107.84	281.24
13	dsm_05m	(0.5, 0.5)	TRUE	493	7740	44441.60
14	dtm_05m	(0.5, 0.5)	TRUE	493	7740	44441.60
15	gcn_conc_PM10_2024	(1000.0, 1000.0)	TRUE	1	1.17	0.03
16	hydrografischekaart_ijsselmeer__hydrografischekaart_ijg_1905	(19, 19)	TRUE	1	7.82	17.44
17	hydrografischekaart_ijsselmeer__hydrografischekaart_ijg_1935	(13, 13)	TRUE	2	17.93	38.92
18	luchtfoto_ecotopen__e_IJsselmeer_2004	(0.5, 0.5)	TRUE	221	3316.68	12810.67
19	luchtfoto_ecotopen__e_IJsselmeer_oost_2011	(0.25, 0.25)	CANCELLED	-	-	-
20	luchtfoto_ecotopen__e_Rijntakken_Oost_2008	(0.25, 0.25)	CANCELLED	-	-	-
21	luchtfoto_lidar__IJssel_2019	(0.05, 0.05)	CANCELLED	-	-	-
22	vaarwegenkaart__vaarwegenkaart	(21, 21)	TRUE	1	12.95	35.60
23	water_en_bodem_ro_raster__veendikte_kaart	(100.0, 100.0)	TRUE	1	4.02	3.19
24	waterstaatskaarten__waterstaatskaarten_5e_editie	(6.3, 6.3)	TRUE	8	98.48	301.17
25	wegenkaart__wegenkaart	(27, 27)	TRUE	1	7.21	16.30
26	wegenkaart__wegenkaart_2013	(21, 21)	TRUE	1	9.63	26.70
27	wegenkaart__wegenkaart_2018	(27, 27)	TRUE	1	7.2	16.30
28	wegenkaart__wegenkaart_2021	(27, 27)	TRUE	1	7.41	16.30

5.4.2. WFS results

From the total of 502 files, 495 were successfully downloaded, 2 failed because could not retrieve total feature count and 5 returned empty results. The successful downloaded files summed up to 10.3GB total and took 4h and 18min to be completed. Examples of these downloaded vectors can be found in **Appendix A4**.

The reason why these 2 records failed remain unresolved. In earlier tests, large part of the downloads failed because the system couldn't retrieve the total number of features from the server. As a result, many records were missed. To improve this, the process was updated to include multiple retry attempts and short pauses between download requests. This greatly increased the number of successful downloads, despite the two cases that still failed.

6. Challenges and limitations

This project encountered several technical and structural limitations related to metadata consistency, service interoperability, and data availability across the *NGR* and associated OGC services.

1. Too general bounding boxes

The bounding boxes provided in the CSW metadata often do not reflect the true spatial coverage of the dataset. Instead, they frequently represent generalized extents (e.g., the entire Netherlands), which complicates download requests, particularly for high-resolution raster datasets where tile-based access is required (**Appendix A5**, example 1).

In some cases, the bounding box accurately reflects the overall extent of the dataset, but the valid data are distributed in a narrow, elongated pattern (e.g., along a river course). This leads the tiling process to include many tiles dominated by missing values, reducing efficiency and increasing processing overhead (**Appendix A5**, example 2). This mismatch can lead to very inefficient requests or even retrieving rasters with only missing values.

2. Inconsistent layer naming across services

There is a lack of consistency between layer names provided in CSW metadata and those listed in *GetCapabilities* documents:

- In some cases, there is inconsistency in naming conventions: some services use double underscores (__) to separate the namespace prefix and layer name, while others use colons (:), as seen in the example "*Carib_N:Caribbean North*" (**Appendix A5**, example 3).
- In other cases, the names used to describe a dataset in the metadata (CSW) do not exactly match the names used in the actual data service (WCS). For example, the CSW metadata might refer to a resource as "*Bathymetrie Nederland Downloadservice*", while the WCS service (in its *GetCapabilities* document) lists it more simply as "*Bathymetrie Nederland*" (**Appendix A5**, example 4). Mismatches like these make it difficult to reliably link the CSW metadata record to the corresponding WCS coverage, especially when the actual coverage name is missing from the CSW metadata. This is problematic because the coverage name is required to construct a valid WCS request.
- Also, cases were found where the CSW metadata uses capital letters while the WCS/WFS services accept only lowercase names. For example, in (**Appendix A5**, example 5), the layer name "*rasters__GEONAM_AKK_BWGSHRST*" must be passed in lowercase to *GetCoverage*, despite appearing in uppercase in *GetCapabilities*.

Such inconsistencies require manual or automated matching logic and reduce the reliability of using metadata records directly to build requests.

3. Axis label inconsistencies in WCS 2.0.1

The *DescribeCoverage* response under WCS version 2.0.1 does not present a standardized format for axis labels. Axis names vary between records and may be labelled as "E/N", "East/North", "X/Y", "x/y". For example, both of the following services, operated by *Rijkswaterstaat*, show different axis naming conventions (**Appendix A5**, examples 6 and 7). These inconsistencies affect how spatial coordinates are interpreted and require careful parsing during data extraction.

4. Incomplete representation of RIVM raster datasets in NGR catalogue

About 6,037 raster layers from RIVM endpoint appear in services (**Appendix A5**, example 8) but only 9 of them present metadata entries in NGR (**Appendix A5**, example 9), while the large majority lack corresponding metadata. This prevents full automation of metadata discovery and complicates traceability and reuse.

5. Differences in WCS/WFS versions and required parameters

Multiple versions of WCS and WFS are in use across providers, each with distinct parameter requirements, and very often this is not specified. This necessitates fallback mechanisms and conditional logic to handle service-specific implementations. Some servers accept `scaleSize`, others require `resx/resy`, and certain responses deviate from expected formats.

6. Limit of data transference per request

Many WCS services impose limits on the amount of data that can be transferred in a single request to prevent excessive memory or CPU usage (Geoserver_Project, n.d.). These limits help ensure server stability but can cause large raster requests to fail unless the data is tiled or the request is adjusted.

7. Limited availability of ecologically relevant data

The *NGR* catalogue contains fewer ecological or high-resolution datasets than initially anticipated. Additionally, the thematic diversity and ecological relevance of available layers were limited. Despite this, parsing metadata remains challenging due to inconsistent structures, terminology, and technical formatting.

8. Non-functional WFS endpoints

Occurrence of metadata records point to non-functional or outdated WFS service URLs (**Appendix A5**, examples 10 and 11). These endpoints fail to return a valid response, making automated data access impossible for that record.

Suggestions for future improvements

The issue of inaccurate or overly generalized bounding boxes could be greatly reduced if the server provided, for each WCS record, a list of tile-specific bounding boxes. These should match the server's limitations and the data's native resolution, and precisely cover the actual area where data is present.

7. Conclusions

This project developed a prototype user-friendly Python workflow to discover, filter, and retrieve geospatial data using OGC services. The pipeline successfully automated processes that typically require significant manual effort, providing more precise spatial and content-based filtering. These results demonstrate the feasibility of using standardized web services to streamline access to ecological data.

In terms of metadata compatibility, 20 out of the 23 selected LTER-LIFE metadata fields were found in NGR records, with 16 exact matches and 4 close matches. Only 3 fields had no equivalent. This indicates that the NGR infrastructure offers foundation to support the metadata requirements of LTER-LIFE.

The main limitation of the current workflow is its execution speed, which is affected by measures such as multiple fallback mechanisms designed to handle imprecise metadata, varying service versions, and inconsistencies across data sources. While these measures improve robustness, they also make the pipeline less practical for large-scale or repeated use. Improving performance will require optimization and possibly the involvement of experts in production-level programming.

Once these improvements are made, the adoption of this workflow within LTER-LIFE could make routine ecological research much more efficient. Tasks that now require several days, such as locating vegetation maps, soil layers, or high-resolution elevation data, could be completed in minutes. Instead of manually downloading and processing multiple datasets, ecologists would have immediate access to ready-to-

use geodata in a consistent format for their study areas. This would allow researchers to dedicate more time to ecological analysis and interpretation and less time to data preparation, ultimately enabling faster, broader, and more ambitious ecological studies.

8. Generative AI statement

Generative AI was employed in this report as a supportive tool. ChatGPT assisted with grammar correction, sentence rephrasing, and drafting Python code. It is important to emphasize that AI served as an aid in the process, not in developing the main ideas and content of the proposal.

9. Code availability

All codes developed and used in this research is publicly available at:

[hudson-passos/research-project-internship-nioo](https://github.com/robert-hudson-passos/research-project-internship-nioo)

The repository includes scripts for data processing, analysis, and visualization, as well as documentation to reproduce the results presented in this thesis. All code is written in Python, and detailed instructions are provided in the README file.

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Appendix A

Appendix A1 | Overview of the 23 LTER-LIFE metadata fields, of which 20 were successfully collected by the pipeline, in accordance with the Dutch Metadata Profile on ISO 19115 for Geography (version 2.1.0).

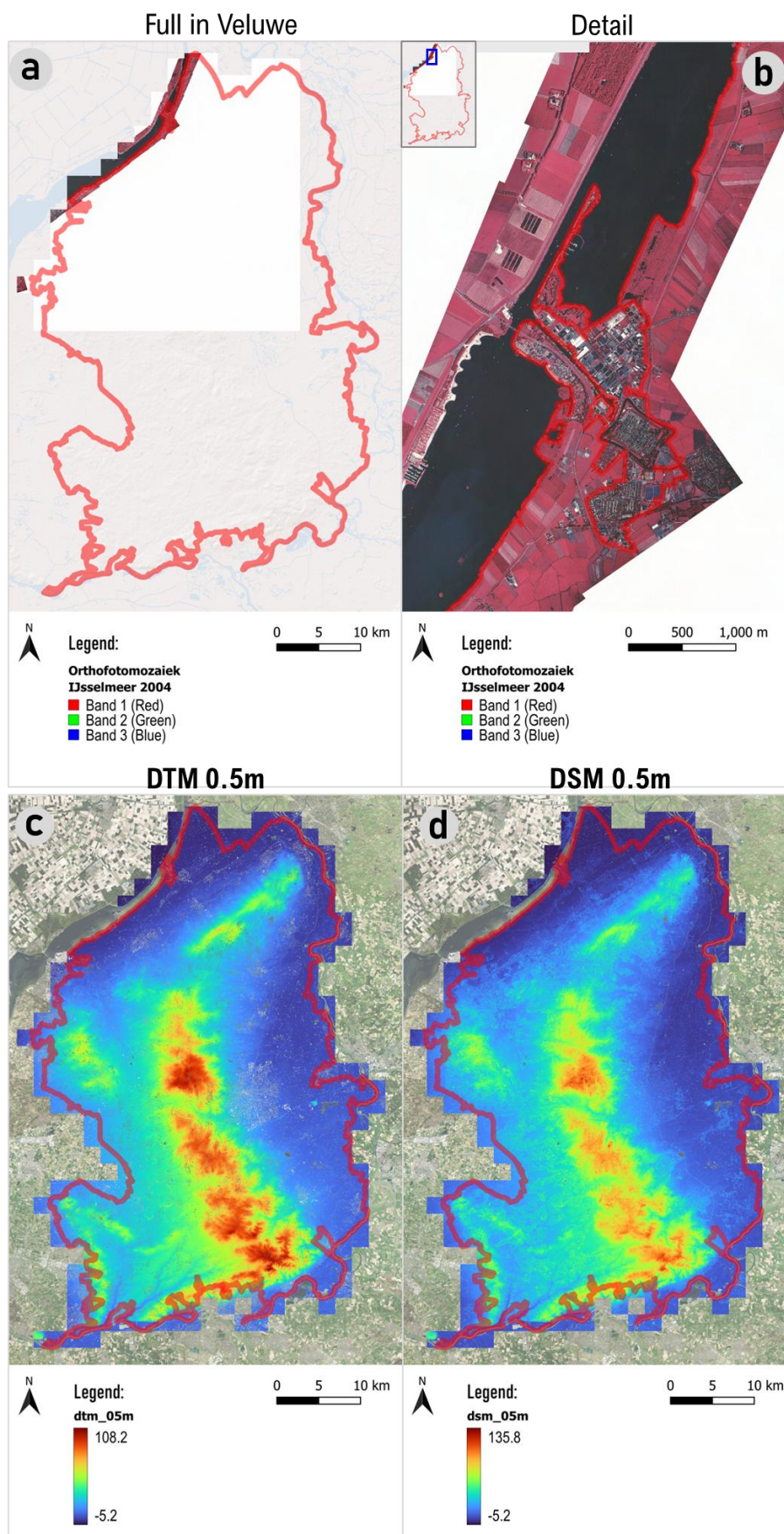
Group	Element	Property (ISO 19115)	Equivalent in LTER-LIFE	Match	Description / Link (ISO 19115)
Metadata on metadata	Metadata date	gmd:dateStamp	gco:date	Exact	The date the metadata was created or modified. The format of the date is YYYY-MM-DD (with dashes). (5.2.43 Metadatadatum)
	Metadata language	gmd:LanguageCode	gmd:language	Exact	The language in which the metadata is described. (5.2.44 Taal van de metadata)
	Responsible party	gmd:contact /.../ gmd:organisationName	gmd:CI_ResponsibleParty	Exact	The name of the organization responsible for the metadata. (5.2.48 Verantwoordelijke organisatie metadata)
	electronicMailAddress	gmd:contact /.../ gmd:electronicMailAddress	gmd:electronicMailAddress	Exact	The email address of the organization responsible for the metadata. (5.2.50 Verantwoordelijke organisatie metadata e mail)
Identification	Landing page	gmd:distributionInfo /.../ gmd:protocol/ gmx:Anchor="landingpage"	dcat:landingPage	Exact	A web page that provides access to the dataset or contains other information. (5.2.30 Protocol; A.19 Codelijst Protocol)
	Title	gmd:identificationInfo/.../gmd:title	dct:title	Exact	This element is used to store the name of the dataset. (5.2.1 Titel van de bron)
	Description	gmd:abstract	dct:description	Exact	Description of the content of the dataset. In this summary, provide user-friendly information about the content of the dataset. (5.2.5 Samenvatting)
	Unique Identifier	gmd:MD_Identifier / gmd:code	dct:identifier	Exact	This element contains the value that uniquely identifies an object. (5.2.4 Unieke Identifier van de bron)
	Resource type	gmd:hierarchyLevel / gmd:MD_ScopeCode / codeListValue	dct:type	Exact	The element indicates what the metadata relates to, the dataset or dataset series. (5.2.46 Hiërarchieniveau)
	Keywords	gmd:keyword	dcat:keyword	Exact	The element contains commonly used words or formalised phrases to describe a dataset or dataset series. (5.2.13 Trefwoord)
Data contact information	Data creator	gmd:pointOfContact /.../ gmd:role / codeListValue="originator"	dct:creator	Exact	This element contains the role of the organization responsible for the metadata: - originator: party that created the data. - pointOfContact: party that acts as a point of contact for exchanging knowledge or providing data. - publisher: party that publishes the data. (5.2.49 Verantwoordelijke organisatie metadata rol; A.3 Codelijst Role)
	Data contact point	gmd:pointOfContact /.../ gmd:role / codeListValue="pointOfContact"	dcat:contactPoint	Exact	
	Data publisher	gmd:pointOfContact /.../ gmd:role / codeListValue="publisher"	dct:publisher	Exact	
Spatial properties	Spatial coverage	gmd:extent/.../gmd:geographicElement	dct:spatial	Exact	This element contains the westernmost, easternmost, northernmost and southernmost coordinates of the dataset. (5.2.25 Omgrenzende rechthoek)
	Spatial resolution	gco:Distance	dct:spatialResolutionMeters	Exact	The element contains the resolution in meters. (5.2.22 Resolutie)
	Spatial reference system	gmd:referenceSystemIdentifier/.../gmd:code	gmd:referenceSystemInfo	Exact	This element contains the Alphanumeric value that indicates the reference system used for the dataset. EPSG issues these codes. For the RD, the code 28992 is used. (5.2.23 Ruimtelijk referentiesysteem)
Temporal properties	Temporal coverage	-	dct:temporal	No match	-
	Temporal resolution	-	dcat:temporalResolution	No match	-
Intellectual rights	License	gmd:resourceConstraints / gmd:MD_LegalConstraints / gmd:otherConstraints	dct:license	Close	This element describes additional legal access and use constraints and must be included when "otherRestrictions" is specified. (5.2.18 Overige beperkingen)
	Access rights	gmd:resourceConstraints /.../ gmd:accessConstraints	dct:accessRights	Close	This element defines legal access restrictions and is mandatory, especially when describing conditions for accessing and using a dataset. (5.2.17 Juridische toegangsrestricties)
Distribution	Distribution access URL	gmd:distributionInfo /.../ gmd:linkage	dcat:accessURL	Close	This element specifies a URL for accessing the dataset and is mandatory if the data is publicly available under the responsibility of the managing organization. (5.2.29 URL)
	Distribution format	gmd:distributionFormat /.../ gmd:name	dct:format	Close	This element contains the name of the distribution format of the dataset. (5.2.26 Naam distributie formaat)
	Distribution byte size	-	dcat:byteSize	No match	-

Appendix A2 | Minimum metadata fields (n=23) used in LTER-LIFE.

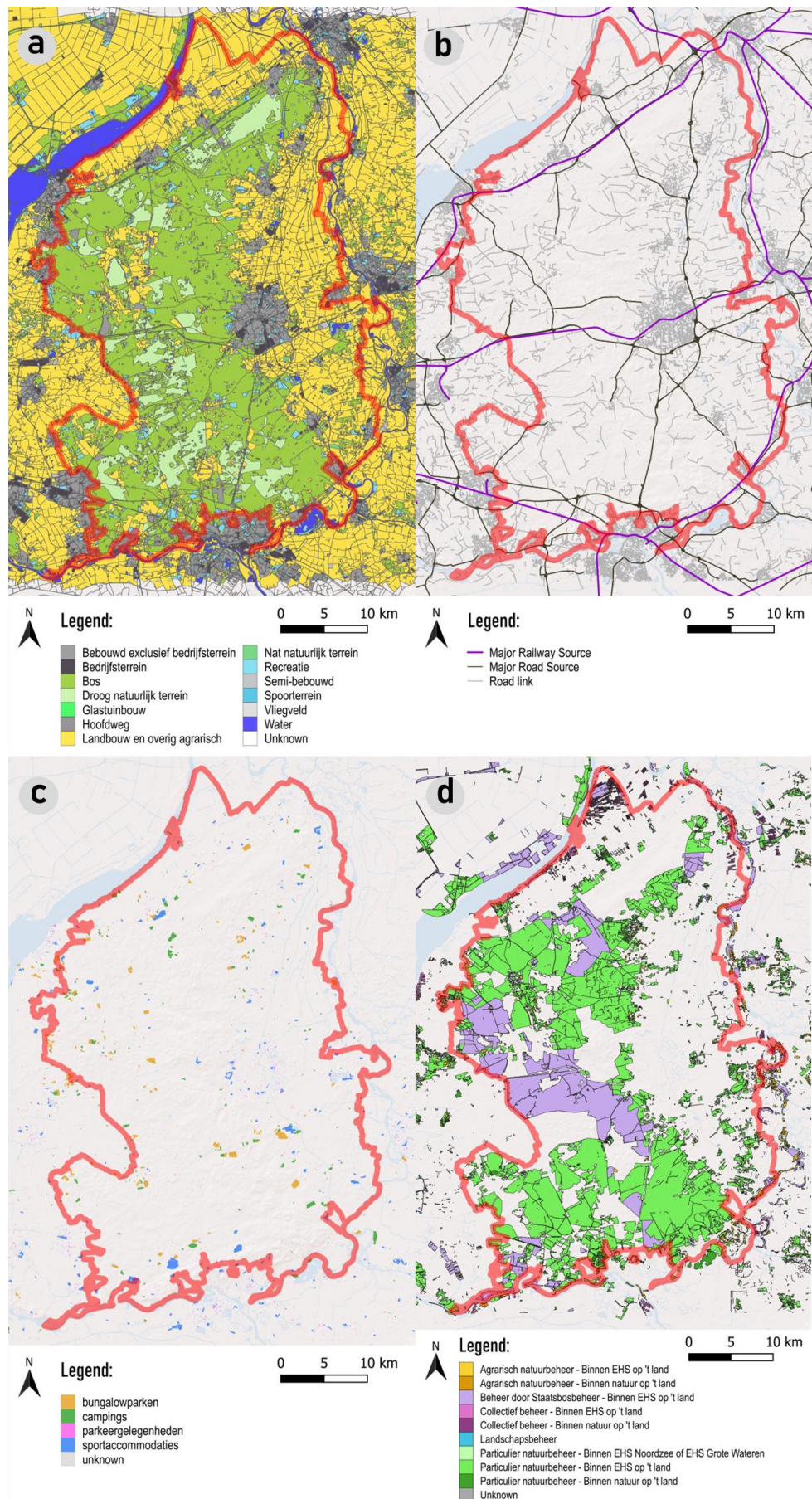
Group	Element	Metadata standard	Property	Instructions
Metadata on metadata	Metadata date	ISO 19115	gco:date	Date specifying when the metadata record was created or updated. The format of the date should be entered as YYYY-MM-DD (with dashes).
	Metadata language	ISO 19115	gmd:language	The language used for documenting metadata.
	Responsible party	ISO 19115	gmd:CI_ResponsibleParty	Responsible party refers to the individual or organization responsible for creating, providing and maintaining the metadata. This could include the name and role of the individual (e.g., data manager), or the name, ROR (Research Organization Registry) ID or website link for the organization.
	ElectronicMailAddress	ISO 19115	gmd:electronicMailAddress	This element contains the email address of the individual or organization responsible for creating, providing, and maintaining the metadata.
Identification	Landing page	DCAT-AP	dcat:landingPage	A web page that provides access to the dataset, its distributions and/or additional information.
	Title	DCAT-AP	dct:title	A name given to the dataset. If no dataset name yet exists, the title could include the type(s) of data collected, the location, and the time period.
	Description	DCAT-AP	dct:description	Description is a free-text account that provides a summary or abstract of the dataset, including its content, purpose, time of collection, and methods used. It can also include a description of the data collected, the location, and the time period.
	Unique Identifier	DCAT-AP	dct:identifier	Identifier is a unique reference to the resource being described or catalogued, such as a Digital Object Identifier (DOI), Uniform Resource Locator (URL), Universally Unique Identifier (UUID), Persistent Uniform Resource Locator (PURL), Archival Resource Key (ARK), Globally Unique Identifier (GUID), or another standardized identifier.
	Resource type	DCAT-AP	dct:type	The nature or genre of the content of the resource. Examples include dataset, software, image, report, model, video, collection, sound, text, service, and workflow, etc.
	Keywords	DCAT-AP	dcat:keyword	Keywords or tags used to describe the content of the data. Multiple entries in the keyword list should be delimited by commas.
Data contact information	Data creator	DCAT-AP	dct:creator	An entity (e.g. person or organisation) responsible for making the resource (i.e. the dataset). This can differ from the responsible party that creates, provides and maintains the metadata.
	Data contact point	DCAT-AP	dcat:contactPoint	Contact information such as name, email address, phone number, and other means of communication for the person or organization responsible for the catalogued resource (i.e. the dataset). This can differ from the responsible party that creates, provides and maintains the metadata.
	Data publisher	DCAT-AP	dct:publisher	The entity responsible for making the resource (i.e. the dataset) available.
Spatial properties	Spatial coverage	DCAT-AP	dct:spatial	Spatial coverage refers to a geographical area where data was collected; a place which is the subject of a collection; or a location which is the focus of an activity. Spatial coverage can be either: a point location, described using geospatial coordinates for a point (e.g. latitude and longitude) an area, described using geospatial coordinates for an area (e.g. a bounding box) a place or region (regional names may be based on legal jurisdiction (e.g. South Australia).
	Spatial resolution	DCAT-AP	dct:spatialResolutionMeters	For raster datasets (images or grids), spatial resolution equals the cell size. For vector datasets, resolution isn't well-defined—just specify the geometry type (polygons, polylines, or points).
	Spatial reference system	ISO 19115	gmd:referenceSystemInfo	The Coordinate Reference System (CRS) defines how spatial data is positioned on Earth using a standardized framework; if your CRS comes from the European Petroleum Survey Group (EPSG) registry, please provide its 4–8 digit EPSG code (you can search for it at http://www.epsg.org). If it is not based on EPSG, please supply its full name and detailed information. Additional help is available at https://spatialreference.org .
Temporal properties	Temporal coverage	DCAT-AP	dct:temporal	Describe the specific time period that a dataset represents, indicating the start and end dates during which the data were collected, generated, or are valid and relevant. The full date range or the range of years could be provided.
	Temporal resolution	DCAT-AP	dcat:temporalResolution	Temporal resolution refers to the discrete resolution of a measurement with respect to the amount of time needed to revisit and acquire data for exactly the same location. It determines how often measurements or samples are taken within a given time period. If the dataset is a time-series this should correspond to the spacing of items in the series. For other kinds of dataset, this property will usually indicate the smallest time difference between items in the dataset.
	License	DCAT-AP	dct:license	Information about the license identifier. A license is a legal document giving official permission to do something with the resource (i.e. the dataset).

Group	Element	Metadata standard	Property	Instructions
Intellectual rights	Access rights	DCAT-AP	dct:accessRights	Information about who can access the resource or an indication of its security status. Embargoed access refers to a resource (i.e. dataset) that has only access to metadata until it is released for open access on a certain date. Embargoes can be required by publisher and funder policies or set by the author (e.g. such as in the case of theses and dissertations). Metadata only access refers to a resource in which access is limited to metadata only. The resource itself is described by the metadata, but the data is not directly available through the system or platform, nor can it be referenced to an open access copy in an external journal or trustworthy archive. Open access refers to a resource that is immediately and permanently online, and free for all on the Web, without financial and technical barriers. The resource is either stored in the repository or referenced to an external journal or trustworthy archive. Restricted access refers to a resource that is available in a system but with some type of restriction for full open access. This type of access can occur in a number of different situations. Some examples are described below: The user must log-in to the system in order to access the resource. The user must send an email to the author or system administrator to access the resource. Access to the resource is restricted to a specific community (e.g. limited to a university community).
Distribution	Distribution access URL	DCAT-AP	dcat:accessURL	Could be any kind of URL that gives access to a distribution of the dataset. E.g. a landing page, download option, feed URL, SPARQL endpoint etc. It can be used for the URL of a service or location that can provide access to this distribution, typically through a Web form, query or API call.
	Distribution format	DCAT-AP	dct:format	Describes the format in which the data will be distributed, including data formats such as CSV, XLS, JSON, etc.
	Distribution byte size	DCAT-AP	dcat:byteSize	The size of the distribution (i.e. dataset) in bytes, kilobytes (KB), megabytes (MB), gigabytes (GB) or terabytes (TB).

Orthofotomozaiek IJsselmeer 2004



Appendix A3 | Additional high-resolution rasters downloaded via WCS: (a) Orthofotomozaiek IJsselmeer 2004 cropped to Veluwe outline; (b) Detail of the Orthofotomozaiek IJsselmeer 2004; (c) DTM, 0.5m resolution; and (d) DSM, 0.5m resolution.



Appendix A4 | Examples of vectors data downloaded via WFS: (a) Bestand Bodemgebruik 2015; (b) three vectors: "Hoofd spoorwegen - Richtlijn Omgevingslawaaï", "Hoofdwegen - Richtlijn omgevingslawaaï", and "Vervoersnetwerken - Wegen - Netwerk"; (c) Kampeervoorzieningen, parkeervoorzieningen en sportaccommodaties voor de samenlevingsatlas; and (d) Natuurmeting Op Kaart 2014: Planologische EHS.

Appendix A5 | Table with links to examples of inconsistencies observed in the metadata and WCS/WFS operations.

Example	Title of the example data	Link with example	Description of the problem
1	Orthofotomozaiek IJssel 2019	Link 1 NGR Website	General bounding box covering the entire country rather than specify the data coverage.
2	Orthofotomozaiek IJsselmeer 2004	Link 2 NGR website WMS	Bounding box covers well but too many areas with missing values to use tiles.
3	Bathymetry Caribbean North	Link 3 GetCapabilities	Check the <wcs:Identifier> element inside <wcs:CoverageSummary> for the coverage name "Carib_N:Caribbean North".
4	Bathymetrie Nederlands deel van de Noordzee dieper dan 10 m LAT - Downloadservice	Link 4 CSW Metadata and the respective Link 4 GetCapabilities	Lack of consistency between layer names provided in CSW metadata and those listed in <i>GetCapabilities</i> documents.
5	Bodemkundige geschiktheid voor akkerbouw	Link example 5	Case where the CSW metadata uses capital letters while the WCS/WFS services accept only lowercase names
6	Bathymetrie versie 2019	Link 6 DescribeCoverage	In <i>DescribeCoverage</i> , the "axisLabels" is "E N".
7	Bodemhoogte kaart, basiskaart voor de Zoute Ecotopen Westerschelde 2020	Link 7 DescribeCoverage	In <i>DescribeCoverage</i> , the "axisLabels" is "X Y".
8	RIVM end point	Link 8 GetCapabilities	RIVM end point showing a large repository of 6,037 raster layers.
9	NGR datasets from RIVM	Link 9 NGR website search	List of 9 rasters datasets from <i>RIVM</i> with metadata in <i>NGR</i> .
10	Fijnstof (PM10) Grootschalige concentratiekaarten Nederland (INSPIRE as-is WCS)	Link 10 GetCapabilities in metadata Link 10 CSW Metadata	This service provided a broken WCS URL link.
11	Roodbaardtuinten	Link 11 URL data provider in metadata Link 11 CSW Metadata	This service provided a broken WFS URL link.